

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCVC305 - SURVEYING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is not a method of plane table surveying?
 - a) Radiation
 - b) Resection
 - c) Traversing
 - d) Intersection of curves
- 2 What is the main principle of surveying?
 - a) Measure from part to whole
 - b) Measure from whole to part
 - c) Measure only elevations
 - d) Measure only distances
- 3 The main advantage of a plane table is:
 - a) Not affected by local attraction
 - b) Plotting and fieldwork are simultaneous
 - c) Can be used at night
 - d) Less number of accessories
- 4 The first step in any engineering survey is:
 - a) Detailed survey
 - b) Preliminary survey
 - c) Reconnaissance survey
 - d) Route survey
- 5 Which instrument is used to set perpendiculars in the field?
 - a) Optical Square
 - b) Compass
 - c) Theodolite
 - d) Dumpy Level

- 6 The sensitivity of a bubble tube depends on:
- Diameter of the level
 - Radius of curvature of the bubble tube
 - Length of the staff
 - Size of the telescope
- 7 In a prismatic compass, the zero of the graduated ring is at:
- North
 - South
 - East
 - West
- 8 Local attraction in compass survey is tested by:
- Measuring only one bearing
 - Comparing fore and back bearings of lines
 - Using theodolite
 - Plotting the traverse directly
- 9 A line of collimation is defined as:
- Horizontal axis of telescope
 - Line joining the optical center and crosshair intersection
 - Line perpendicular to the bubble tube
 - Curved level line
- 10 Reciprocal leveling is used when:
- Two points are at the same level
 - Distance between stations is long
 - Line of sight is obstructed
 - Instrument cannot be set up midway
- 11 Which of the following is not required in preliminary surveys?
- Soil testing
 - Cross-sectional leveling
 - Curve setting
 - Collection of topographic data
- 12 Which table is used for balancing a closed traverse in theodolite surveying?
- Traverse table
 - Gale's table
 - Contour table
 - Height table
- Q2. Solve the following.
- Explain reciprocal ranging with neat sketch. 6
 - Explain in detail classification of survey. 6
- Q3. Solve the following.
- List various methods of plain table surveying and explain any two in detail. 6
 - Find the included angles between the lines OP and OQ whose reduced bearings are as given below. 6
 - N 21°30' E and S 65°W.
 - N 32° W and S 51°45' W.
- Q4. Solve Any Two of the following.

- A) The following are the consecutive readings on a 4 m leveling staff on a continuously sloping ground at an interval of 30m
0.880, 1.600, 1.970, 2.550, 2.990, 3.485, 1.250, 1.980, 2.465, 3.740, 0.920, 1.145, 1.850, 2.740.
The RL of first point is 200.00m. Rule out a page of level field book and enter the above readings. Calculate the reduced levels of all the points by rise and fall method. Also find the gradient of the line joining the first and the last point. 6

- B) With neat sketches explain the characteristics of contours. 6

- C) Explain profile levelling and cross sectioning. 6

Q5. Solve Any Two of the following.

- A) Define the following terms: 6

a. Swinging

b. Face right

c. Face left

d. Transiting

e. Plunging

f. Transit Theodolite

- B) State and explain sources of error in theodolite work. 6

- C) Explain the permanent adjustment of theodolite. 6

Q6. Solve Any Two of the following.

- A) Write a short note on mine surveying. 6

- B) Enlist important activities in survey for construction of a tunnel and explain in brief any one of it. 6

- C) Enumerate the various types of horizontal curves and explain each in brief. 6

*** End ***